

Racial Bias in COMPAS

Introdução à Aprendizagem Automática – DETI
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What is COMPAS?

Essentially, it's a software who assigns a prediction score, given prior to legal judgment, based on the defendant's background and other data.

COMPAS = 1 → low chance...

COMPAS = 10 → **high chance!**

Correctional
Offender
Management
Profiling for
Alternative
Sanctions

When is it used?

In the United States, judges use it to help them make a decision on sentences.



What's the problem?

It helps decide someone's freedom.

If it's wrong, innocent people suffer in jail.

Multiple studies have exposed racial bias in COMPAS...



What did we do?



- 01 We took ProPublica's Broward County dataset;
- 02 We reestimated the COMPAS scores with our chosen models;
- 03 We analyzed racial bias within the algorithm's performance;
- 04 We studied a model for recidivism prediction, beyond COMPAS, with the same dataset to compare behaviors.

The Dataset



The Dataset

From ProPublica's COMPAS analysis.

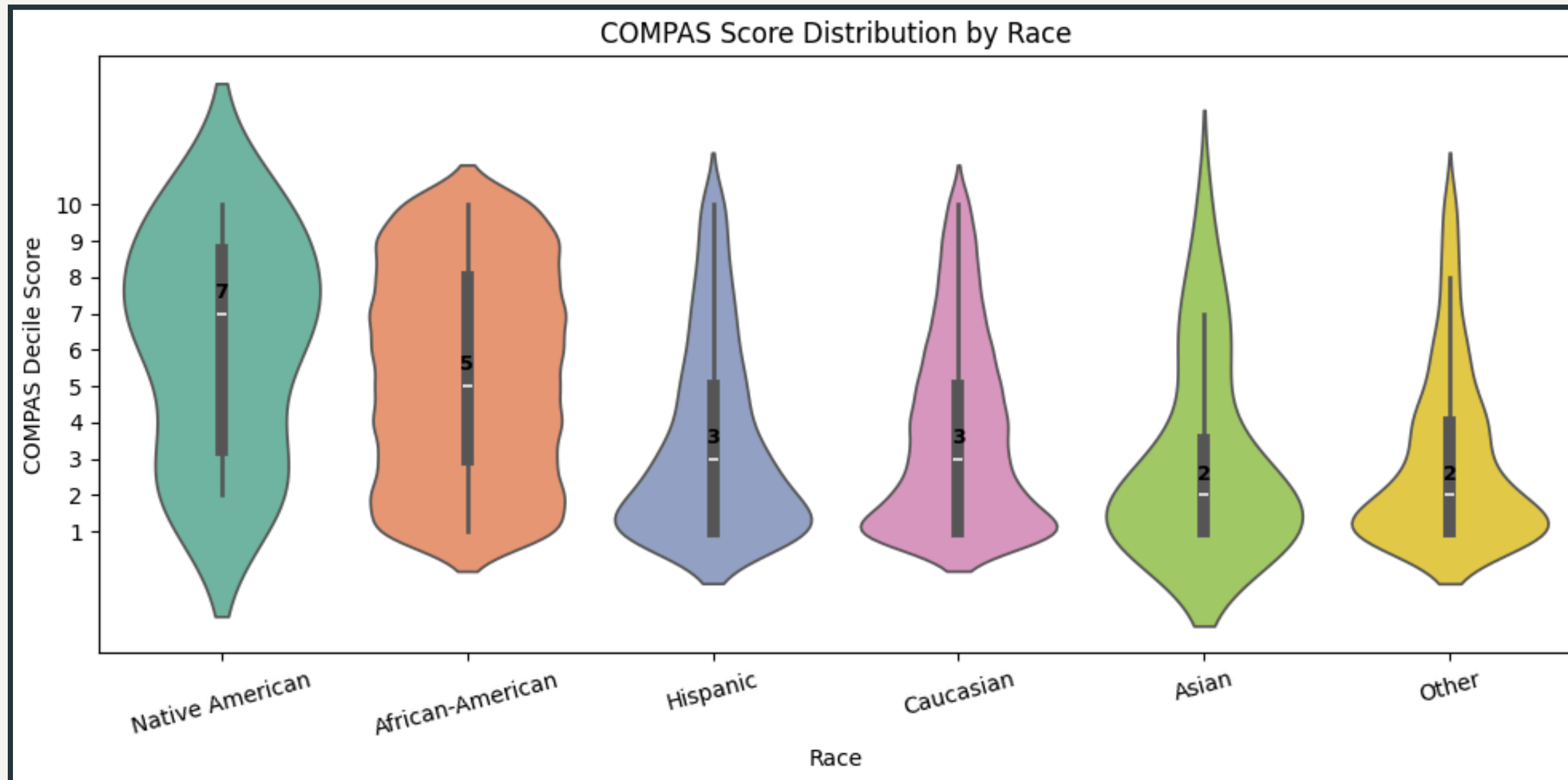
53 columns. Contains:

- Personal attributes;
- Criminal background;
- Current offense information;
- Final scores and recidivism outcomes.

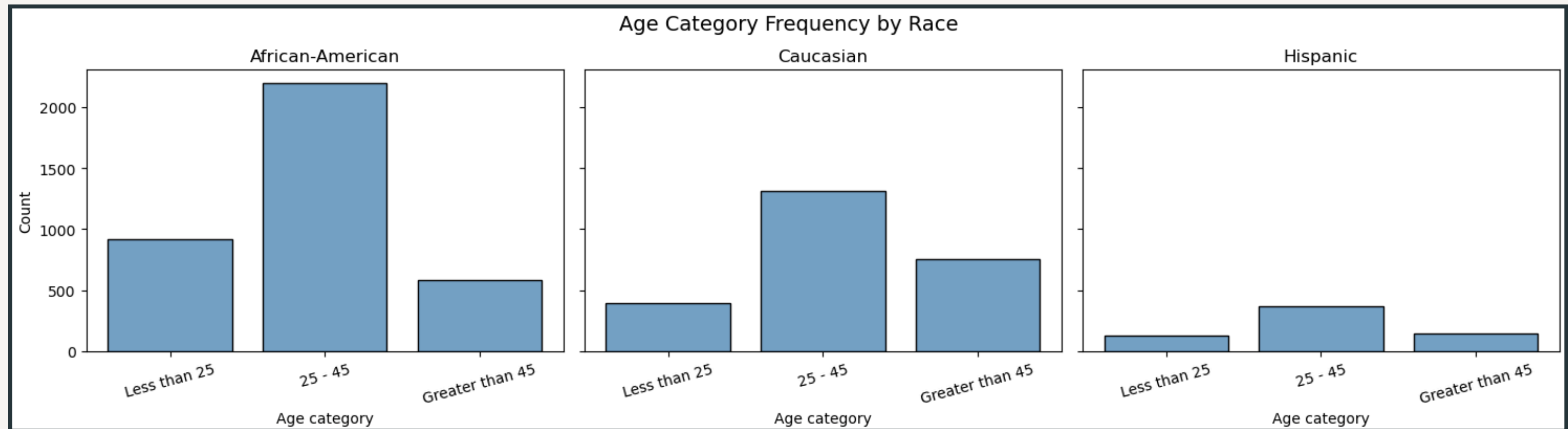
We then further processed the data to better fit the study case.



Decile Score per Race Group



Age Distribution per Race Group



Data Preprocessing

Removed future columns.

Feature Engineering:

- age_at_event;
- priors_per_adult_year;
- first_offense;

Categorical values → Boolean flags.

Final dataset: 21 features, 6172 records.



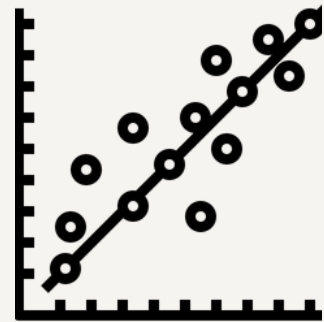
Modeling



Models Used

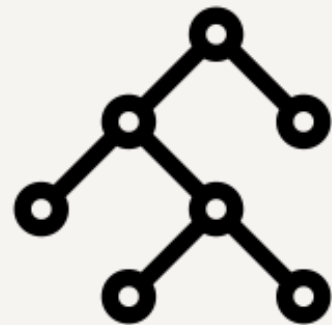
Decile Score Prediction

01



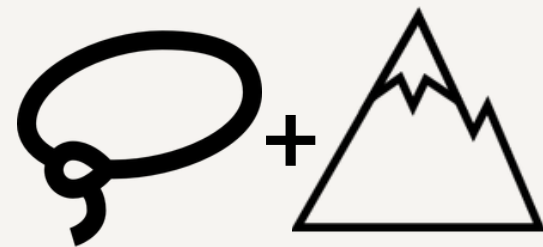
QuantileRegressor: Used to predict conditional quantiles rather than the mean, making the score predictions robust to outliers.

02



RandomForestRegressor: For capturing complex, non-linear patterns in the scoring data.

03



ElasticNet + Polynomial: A linear model combining Lasso and Ridge regularization to prevent overfitting.

Recidivism Prediction

04



RandomForestClassifier: Handles non-linear relationships well.

Training - Decile Score

Hyper parameter tuning (GridSearchCV, 5 folds, R2 optimized)

Number of estimators
Max Depth
Min Samples

RandomForestRegressor

Polinomial Factor
Alpha (Ridge)
L1 Ratio (Lasso)

ElasticNet + Polynomial
Features

Polinomial Factor
Alpha
Quantile

QuantileRegressor

Split

80/20 - Train/Test
Stratezigid

Scaling

StandardScaling

Training - Recedivism

Hyper parameter tuning (GridSearchCV, 5 folds, F1 optimized)

Number of estimators

Max Depth

Min Samples

RandomForestClassifier

Split

80/20 - Train/Test

Stratezigned

Scaling

StandardScaling

Results

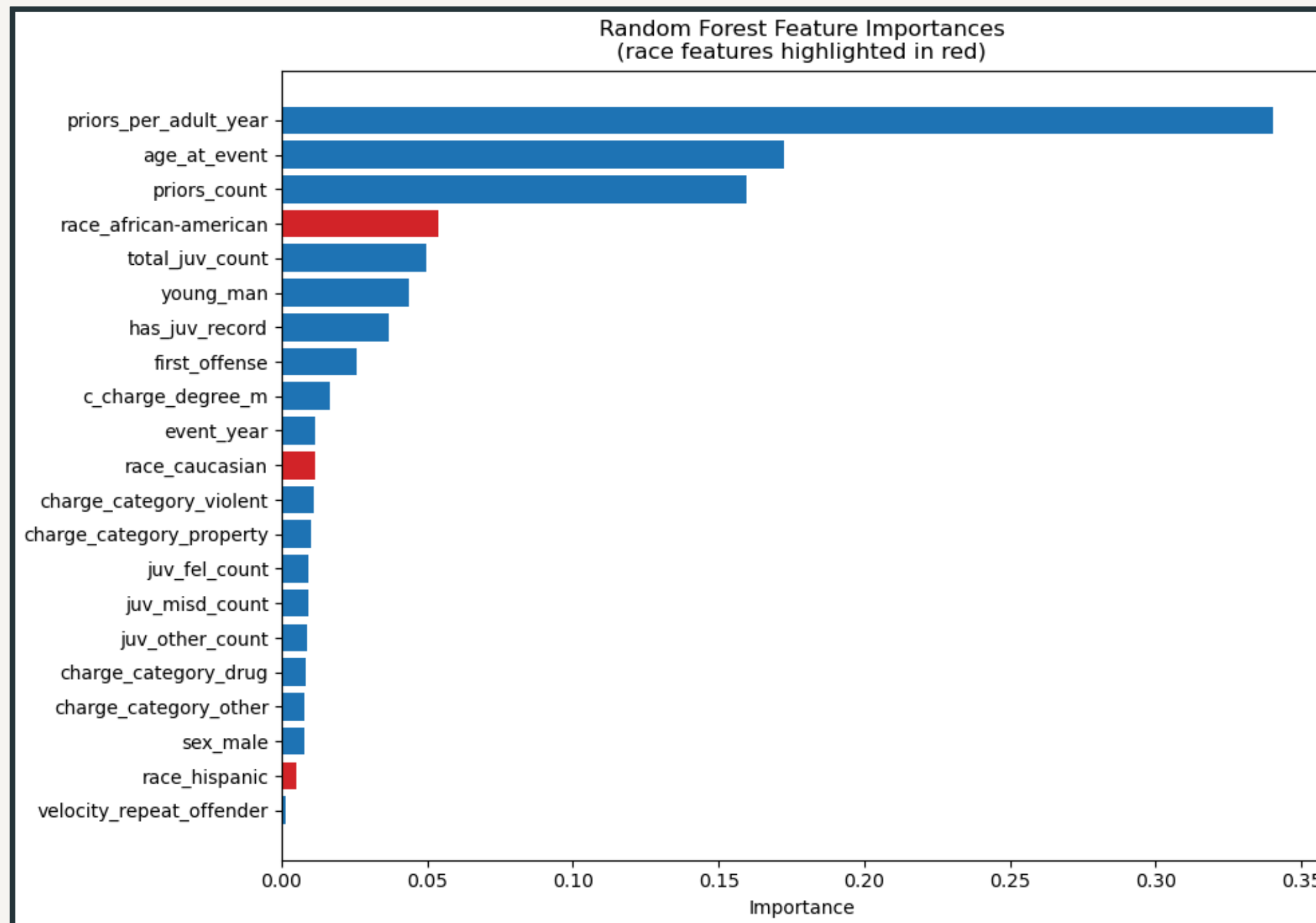


Performance Plateaued

	COMPAS	Recidivism Predictor
Race-Aware	$R^2 \simeq 0.5$	$\text{Acc} \simeq 0.7$
Race-Blind	$R^2 \simeq 0.5$	$\text{Acc} \simeq 0.7$

Racial features seemed to have no importance, so we dug a little deeper...

Race Importance (COMPAS)

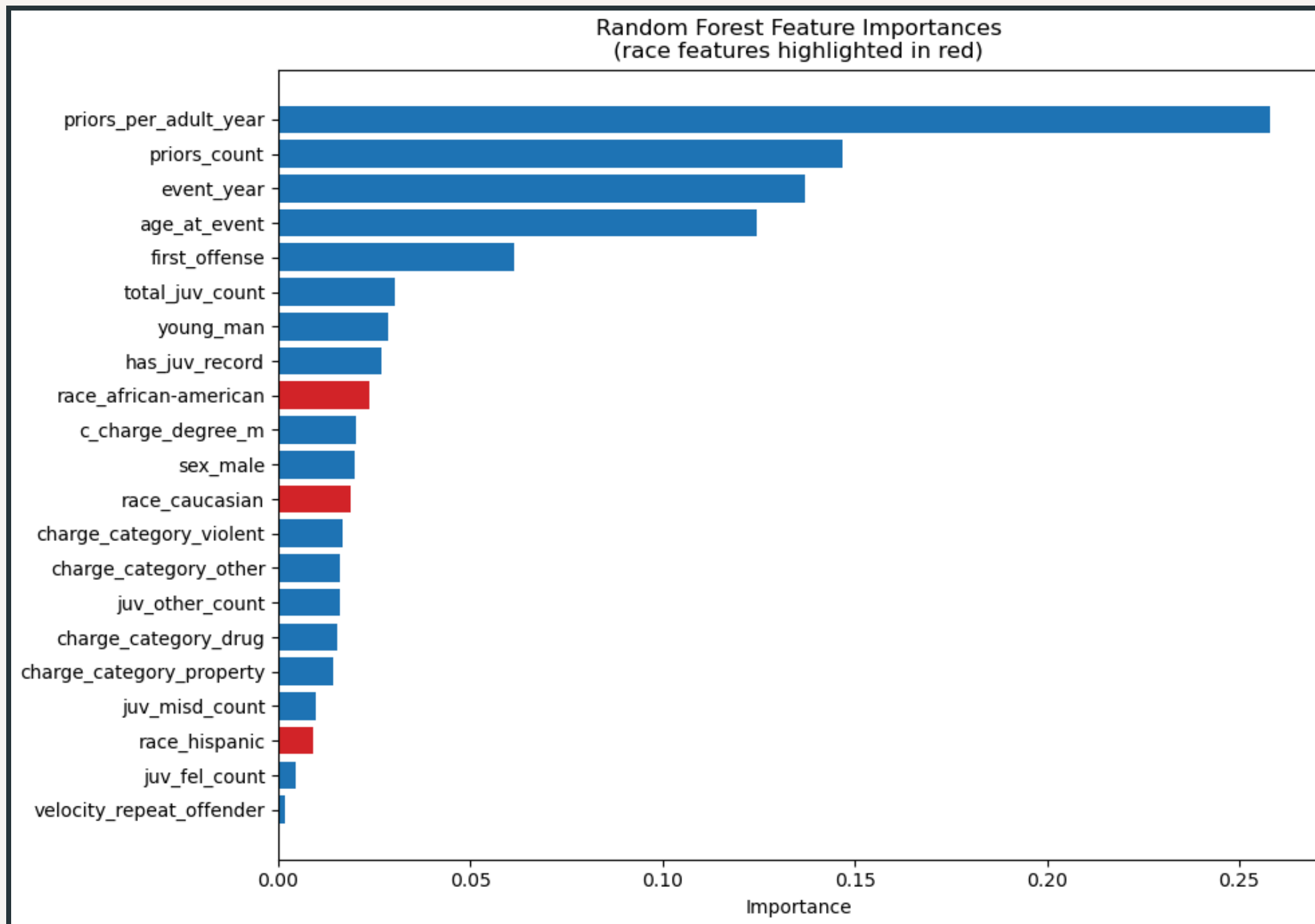


Being African-American ranks 4th .

Being Caucasian only ranks 11th .

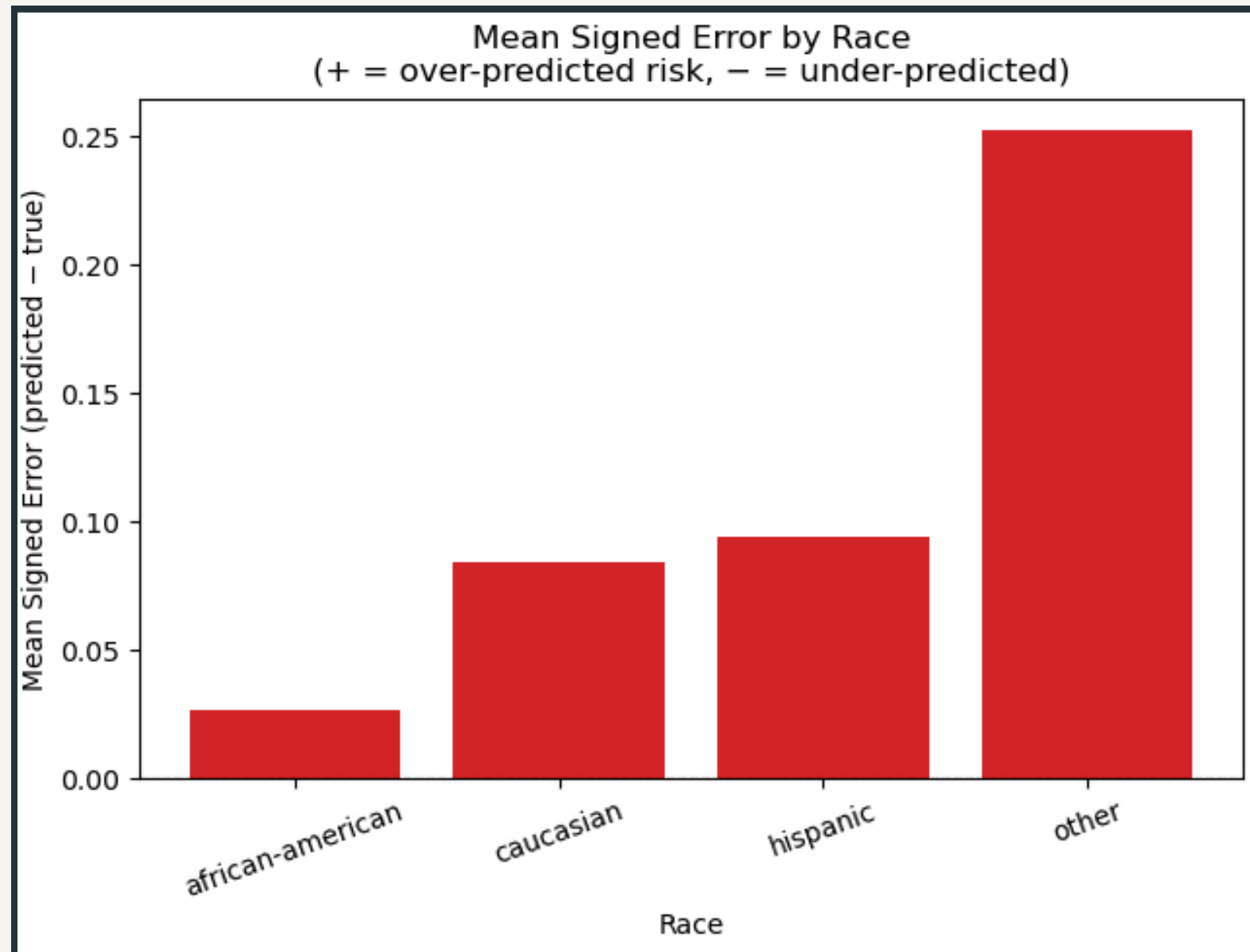
A 5x difference.

Race Importance (Recidivism Class.)

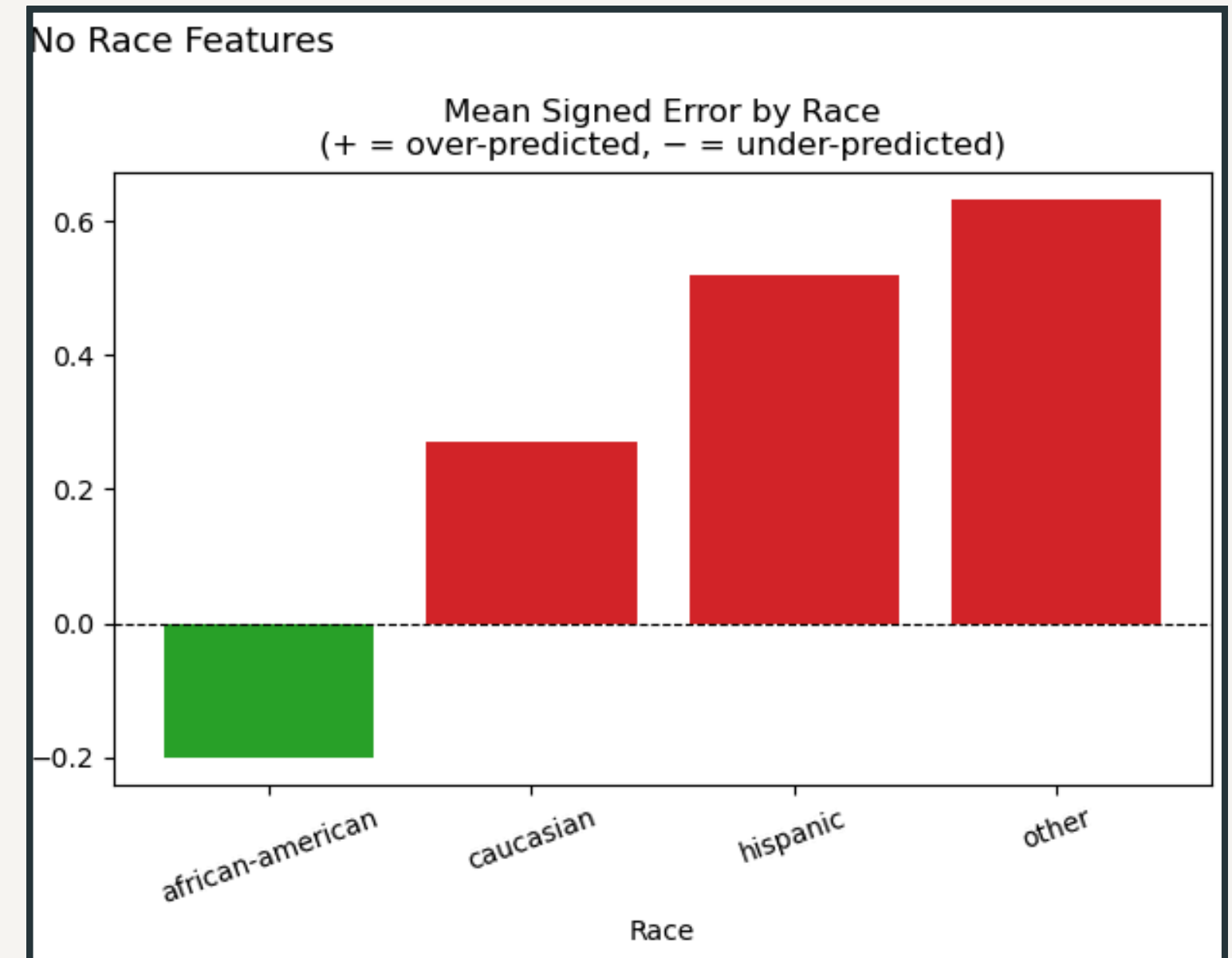


Here the disparity is much smaller, as is the importance.

Mean Signed Error (COMPAS)

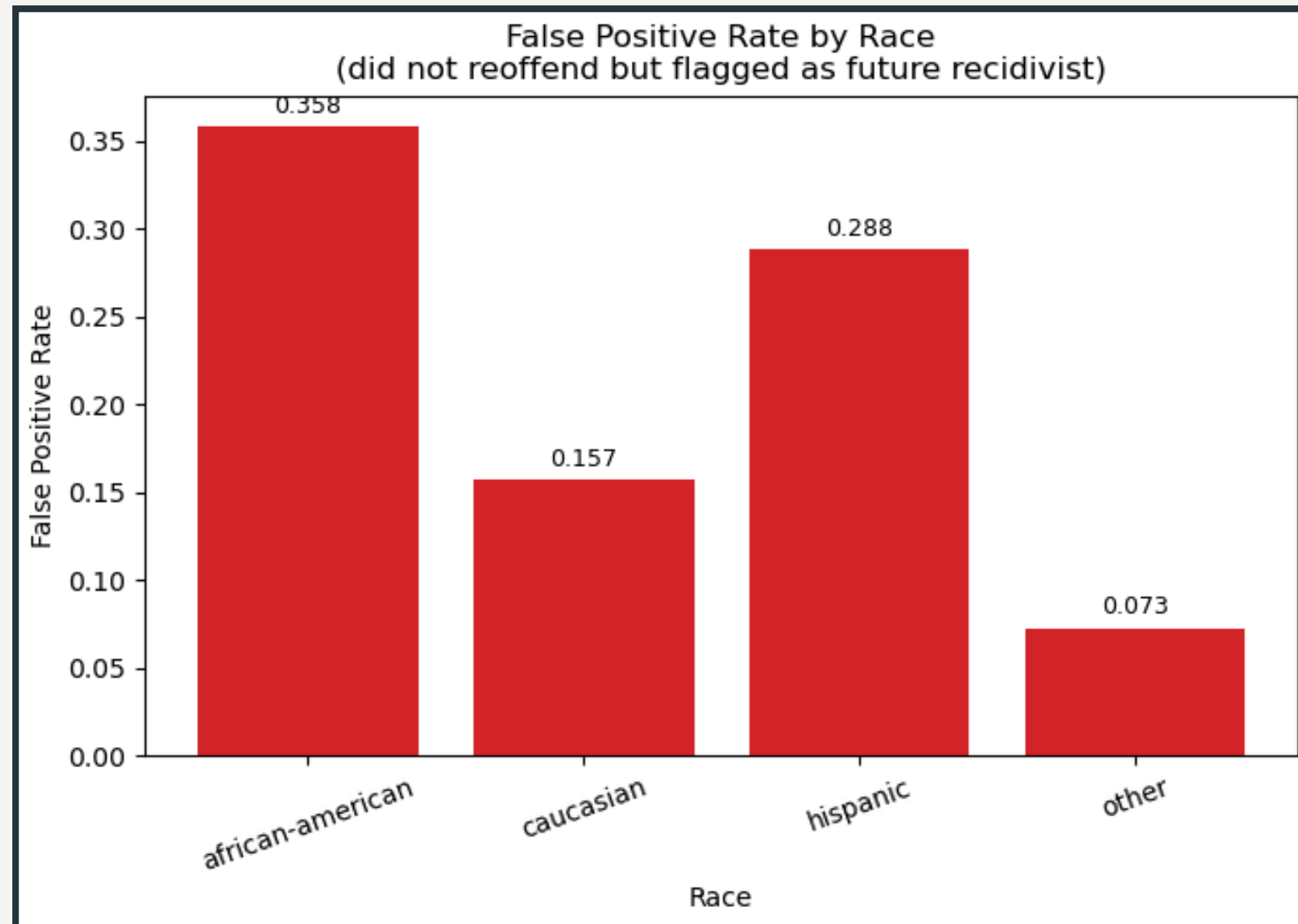


Race Aware

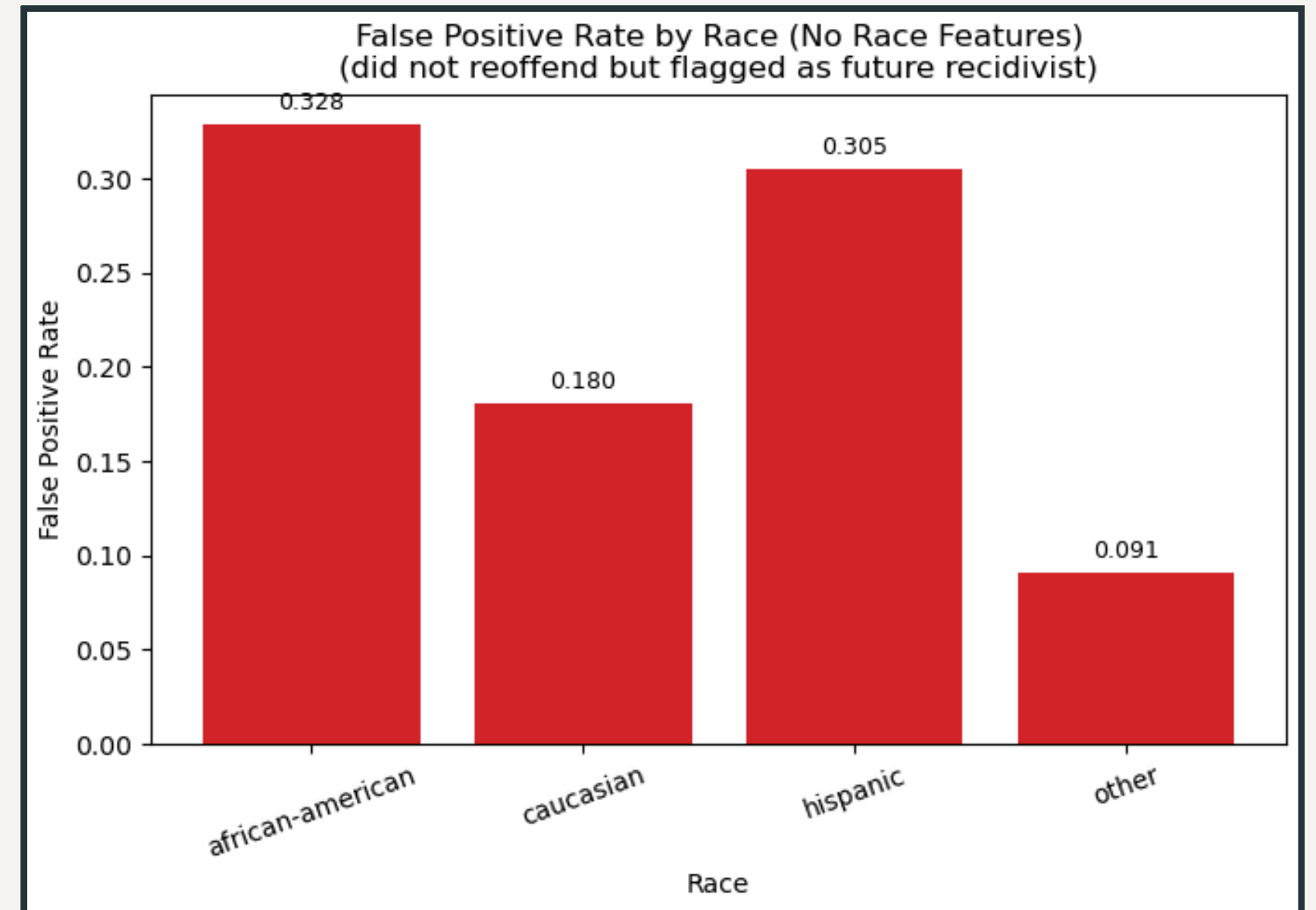


Race Blind

FPR by Race (Recidivism Class.)



Race Aware



Race Blind

Conclusions



Conclusions

- 01 Race does indeed matter in COMPAS.
- 02 It acts as a corrective value.
- 03 The bias persists beyond COMPAS.

How can we fix it?

We can't.

It's a systemic problem,
not a technical one.



Why?

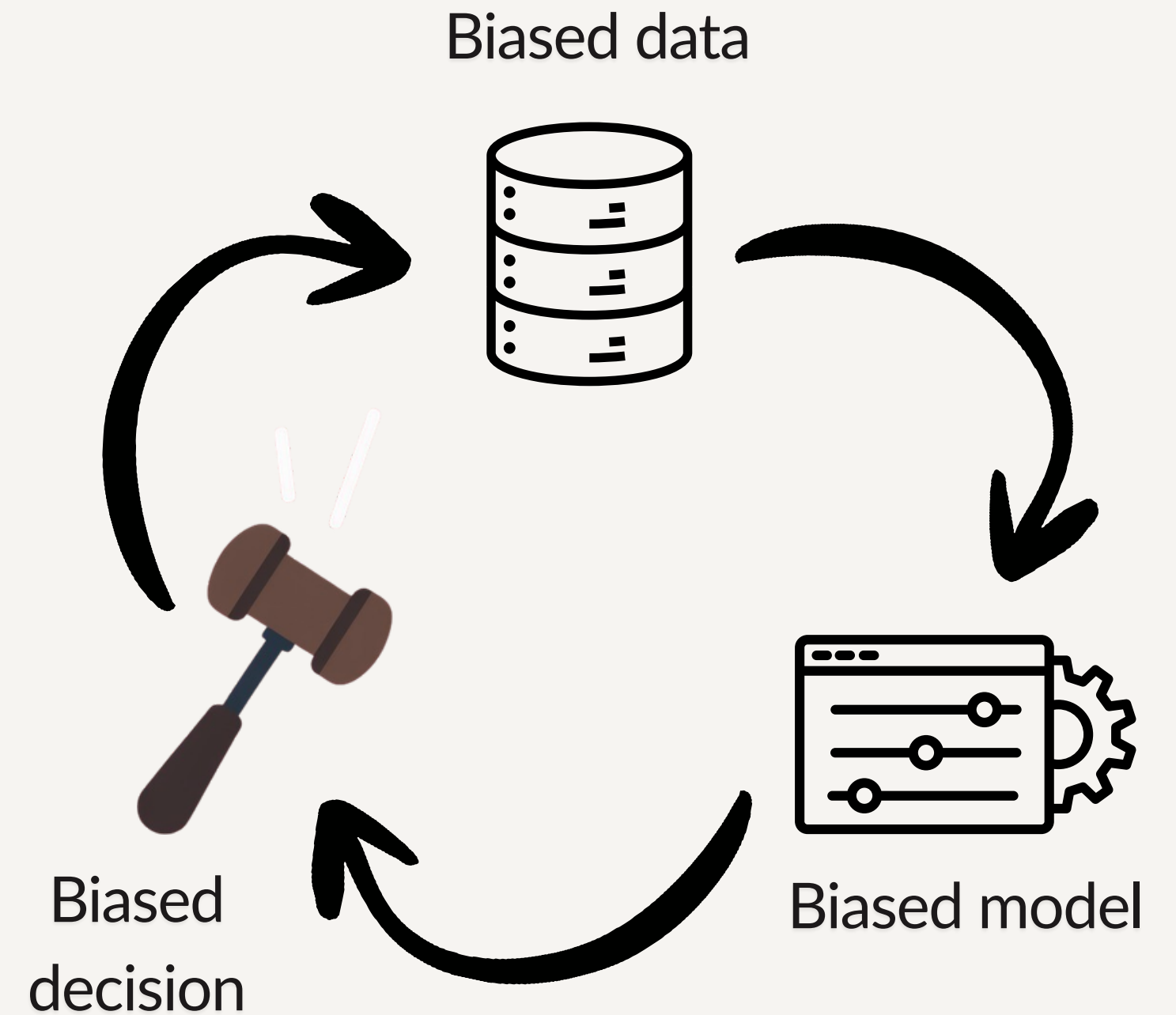
- 01 Even when race is removed, it finds its way through proxy features.
- 02 Bias persists in the data, not in the models.
- 03 The existence of the data in itself carries bias.

Why it's scary...

Not only does it have bias,
it **institutionalizes** it.

It creates a **feedback loop** where biased data creates biased decisions, thus creating more bias.

Instead of fighting **social inequalities**, it **exacerbates** them.



Possible solutions

- Be aware of bias in these tools;
- Use datasets with more even representation.



Thank You

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